

Michael Tin

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EDUCATION

University of California, Riverside (UCR)

Master of Science in Computer Engineering

Apr 2026 – Apr 2028

Riverside, CA

University of California, Riverside (UCR)

Bachelor of Science in Computer Engineering

Sep 2021 – Jun 2025

Riverside, CA

EXPERIENCE

Software Engineering Intern

National Aeronautics and Space Administration (NASA)

Aug 2026 – Present

Hampton, VA

- Develop a comprehensive framework for modernizing legacy PHP code in the NASA Earth Science Strategic Integration Environment (NESSIE) for PHPUnit test and evaluation targeting 100% code coverage.
- Establish a robust CI/CD pipeline and technical documentation within the Flight Systems Software Branch to standardize automated test execution, ensuring the reliability of mission-critical data for Earth science portfolios.

Research Assistant (UIUC SRSE 2026)

University of Illinois Urbana-Champaign (UIUC)

Jun 2026 – Present

Remote

- Develop core framework components for AEBench, a benchmark for evaluating AI agents on artifact evaluation tasks, building a shared runtime to execute versioned research cases and record reproducible results.
- Engineer automated Python evaluation pipelines that compare agent outputs against reference data by utilizing statistical analysis methods, producing objective technical assessments of agent performance.

Technology Director

The Highlander Newspaper

Sep 2022 – Jun 2025

Riverside, CA

- Engineered hardware and network infrastructure for UCR's student-run newspaper, providing on-demand technical support and workflow training to ensure minimal downtime during weekly production.
- Devised a website optimization strategy centered on content delivery optimization, reducing website latency by 82.8% and growing readership by 73.3% to 40,000 monthly average users (MAUs).

PROJECTS

Scansheet | Rust, Tauri, JavaScript, React

Oct 2025 – Jun 2026

- Developed a full-stack attendance management system that eliminates manual entry errors through an automated QR code scanning infrastructure, reducing administrative overhead in educational settings.
- Engineered comprehensive data portability tools to streamline data management, including bulk CSV imports, date/category filtering, customizable PDF reports, and automated badge generation for users.

XV6 Kernel Optimization | C, Assembly, RISC-V

Jan 2025 – Mar 2025

- Engineered advanced system calls and symbolic links for the XV6 kernel, expanding core functionality with file system traversal, inter-process communication, and optimized I/O operations.
- Implemented dynamic page fault management techniques, including lazy page allocation, Copy-on-Write (COW) fork, and memory mapping, to streamline program execution while minimizing system resource consumption.

Integrated Arcade System | C, C++, UART, SPI, AVR

Oct 2024 – Dec 2024

- Engineered a dual-microcontroller system for "Connect 4" and "Whack-a-Mole" using ATmega328P and FRDM-K64F, synchronizing logic via UART/SPI to drive score displays, sound effects, and background music.
- Developed low-level drivers for an ST7735 TFT display, 74HC595 shift registers, and ULN2003 stepper motors, optimizing ISRs to handle asynchronous player inputs while managing game states with physical feedback.

SKILLS

Languages: Python, Java, C, C++, C#, Rust, JavaScript, TypeScript, SQL, MATLAB, Verilog, VHDL, Assembly

Frameworks: React, React Native, Next.js, Express.js, Angular, Flutter, Electron, Tauri, PyTorch, TensorFlow

Developer Tools: Git, GitHub Actions, Docker, Kubernetes, PostgreSQL, MongoDB, WSL, CMake, GDB, QEMU

EDA and Simulation Tools: PSpice, Capture CIS, Vivado, ModelSim, Synopsys Custom Compiler, GTKWave